



24EEE432 Cyber Physical System for Industrial Applications

Smart Parking Infrastructure Using Cyber-Physical Systems

Report

Submitted by

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Amrita School of Engineering, Coimbatore
Department of Electrical and Electronics Engineering

Vision

Mould generations of electrical and electronics engineers on global standards with multi-disciplinary perspective to meet evolving societal needs.

Mission

- Empower students with knowledge in Electrical, Electronics and allied Engineering facilitated in innovative classrooms and state-of-the art laboratories.
- Inculcate technical competence and promote research through industry interactions, field exposures and global collaborations.
- Promote professional ethics and selfless service.

Program Educational Objectives (PEOs)

PEO1: Graduate can demonstrate electrical and electronics engineering problem solving skill along with proficiency in communication and professional excellence in project management and execution.

PEO2: Graduate can be employable in engineering services including ICT enabled sectors and also motivated for entrepreneurship.

PEO3: Graduate will be competent for higher studies in world class universities and research in industrial organizations.

PEO4: Graduate will manifest social commitment, environmental awareness and moral and ethical values in professional and other discourses.

Program Specific Outcome (PSOs)

PSO1: Build and manage electro dynamic systems using Knowledge on electrical technology and semiconductor devices for allied services.

PSO2: Use computational tools and network dynamics for design, analysis and control of power systems integrated with renewable energy and Electric Vehicle.

PSO3: Leverage digital technologies employing state-of-the art control techniques and embedded controllers for industrial applications.

Course Objective:

- Understand and develop hardware and software design requirements of Cyber Physical System for Engineering Applications from Embedded Perspective
- Design and implement a prototype of Embedded Cyber Physical Systems using ARM processor and programming skills in Embedded C, Micro python
- Develop embedded projects for Smart City, Smart Civil Structures, Health Care Application, Automotive Mechatronics, Hybrid Electric Vehicles, Robots for Industrial Automation

Course Outcomes:

- CO1:** Program for ARM Processor STM32F103 in C language using STM Cube IDE Software Development Environment
- CO2:** Gain skills for interfacing ARM processor STM32F103 with sensors like Ultrasonic sensor, Hall effect sensor, Tilt Sensor, actuators like servo Motors and displays like OLED and LCD Displays
- CO3:** Program Raspberry Pi Pico using the Micropython language using Thonny IDE Software Development Environment and develop simple embedded system prototypes using Raspberry Pi Pico to interface with sensors, OLED and LCD displays, motor
- CO4:** Develop Applications of CPS from embedded perspective in Automotive Mechatronics and Healthcare Applications
- CO5:** Develop Applications of CPS from embedded perspective in Hybrid Electric Vehicle, Industrial Motor Control and Robotics

CO-PO Mapping

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO														
CO1	3	3			3							1	2	3
CO2	3	3										1	2	3
CO3	3	3			3							1	2	3
CO4	3	3	3	2	3		2	2	2	2	1		2	3
CO5	3	3	3	2	3		2	2	2	2	1		2	3

PO1-Engineering Knowledge, PO2-Problem Analysis, PO3-Design/Development of Solutions, PO4-Conduct Investigations of Complex Problems, PO5-Modern Tool Usage, PO6-The Engineer and Society, PO7-Environment and Sustainability, PO8-Ethics, PO9-Individual and Team Work, PO10-Communication, PO11-Project Management and Finance.

Washington Accord Knowledge Profiles (WK) mapping	Strongly Supports POs	Justification
WK2 (Mathematics & Computation)	PO1, PO2	Embedded logic, timing, and control algorithms
WK3 (Engineering Fundamentals)	PO1, PO2, PO3	ARM architecture, sensors, actuators
WK4 (Engineering Specialization)	PO1, PO2, PO4	CPS, Embedded Systems, STM32, Pico
WK5 (Engineering Design)	PO3, PO5	CPS application design & prototyping
WK6 (Engineering Practice)	PO4, PO11	IDEs, debugging, and hardware interfacing
WK7 (Engineering Tools)	PO5	STM Cube IDE, Thonny, MicroPython
WK8 (Ethics, Environment & Sustainability)	PO6, PO7	Healthcare, EV, safety-critical CPS
WK9 (Individual & Team Work)	PO9, PO10	Projects-CPS applications



BONAFIDE CERTIFICATE

This is to certify that the cyber-physical system project report entitled
“Smart Parking Infrastructure Using Cyber-Physical Systems”

submitted by

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**Associate Professor, Department of Electrical and Electronics, School of Engineering,
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is in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology**
in **“Electrical and Electronics Engineering”** is a bonafide record of the work carried out at Amrita
School of Engineering, Coimbatore.

Internal Examiner

External Examiner

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1. Aim

The aim of this project is to design and implement a Cyber-Physical System (CPS) based Smart Parking and Access Control System that integrates real-time sensing, embedded computation, automated control, and feedback mechanisms to optimize parking space utilization, enhance security, and support sustainable urban infrastructure development.

The system aims to develop an intelligent parking management framework capable of detecting vehicle presence, managing slot availability, controlling gate access through automated actuation, and providing real-time monitoring through display and wireless communication modules. The proposed system emphasizes reliability, scalability, energy efficiency, and sustainable operation aligned with global development objectives.

2. Problem Statement

Urban areas are experiencing a rapid increase in the number of vehicles, leading to significant challenges in parking space management, traffic congestion, fuel wastage, and security concerns. Conventional parking systems rely heavily on manual monitoring and lack real-time information regarding slot availability, resulting in inefficient space utilization and increased carbon emissions due to unnecessary vehicle movement. Additionally, traditional systems provide limited security mechanisms and minimal integration between sensing, monitoring, and control processes.

Therefore, there is a need for an intelligent and automated parking management solution that integrates real-time sensing, embedded processing, and automated access control to enhance operational efficiency, improve security, and support sustainable urban infrastructure. The proposed Cyber-Physical Smart Parking and Access Control System addresses these challenges by enabling real-time vehicle detection, automated gate management, parking slot monitoring, and wireless status communication within a unified embedded framework.

3. Proposed Solution

To address the challenges associated with conventional parking systems, the proposed solution is the development of a Cyber-Physical System (CPS) based Smart Parking and Access Control System that integrates sensing, embedded processing, automated control, and communication within a unified framework. The system utilizes ultrasonic sensors to detect vehicle presence and monitor parking slot occupancy in real time. Additional sensors such as PIR, hall effect, tilt, sound, and touch modules enhance security monitoring and access management.

The sensed data is processed by embedded microcontrollers, which execute decision-making algorithms to determine parking availability and authorize gate operation. Based on system logic, a stepper motor actuates the parking gate, enabling controlled entry and exit of vehicles. Real-time status information, including slot availability and security alerts, is displayed through OLED, and LED matrix modules. Wireless communication via WiFi interface allows remote monitoring and system supervision.

The proposed architecture ensures efficient parking space utilization, improved security, reduced traffic congestion, and enhanced operational reliability. The modular design allows scalability for integration into larger smart city infrastructures while maintaining energy-efficient and sustainable system performance.

4. Key Features of the Solution

The key features of the proposed Cyber-Physical Smart Parking and Access Control System are as follows:

- Real-time vehicle detection and parking slot occupancy monitoring using ultrasonic sensing technology.
- Automated gate access management through embedded microcontroller-based control and motor actuation.
- Intelligent parking availability computation for optimized space utilization and reduced congestion.
- Integrated multi-sensor security monitoring using PIR, Ultrasonic and IR sensors.
- Real-time system status and alert display through OLED, and LED matrix interfaces.
- Wireless communication capability for remote monitoring and supervisory control.
- Autonomous decision-making framework enabling reliable and efficient system operation.
- Modular, scalable, and energy-efficient architecture suitable for smart city infrastructure integration.

5. Project Design and Architecture

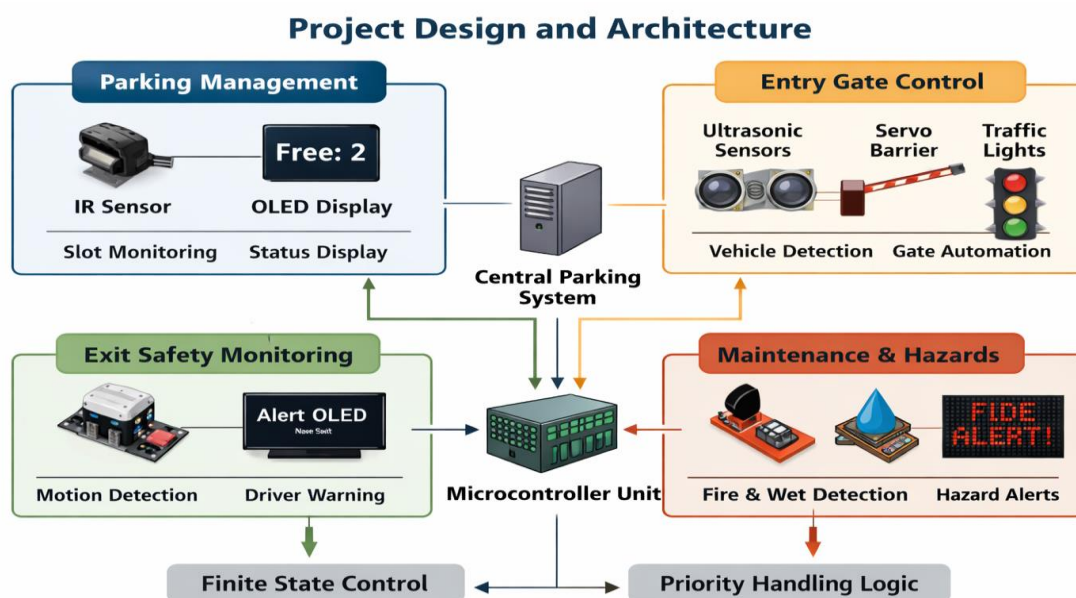


Fig.1 Project Architecture

The proposed smart parking system is designed using a modular and distributed architecture integrating embedded control units with a cloud-based monitoring platform. The overall design follows a layered structure consisting of sensing, control, actuation, and cloud interface layers to ensure scalability, reliability, and safety.

A. Architectural Overview

The system comprises four primary subsystems interconnected through a central microcontroller unit:

1. Parking Management Module
2. Entry Gate Control Module
3. Exit Safety Monitoring Module
4. Maintenance and Hazard Detection Module

Each module operates independently while communicating status information to the central controller. A cloud API and web dashboard interface enable remote monitoring and slot management

B. Parking Management Module

This module performs real-time parking slot monitoring using infrared sensors. Each slot is categorized into three logical states: Free, Reserved, or Occupied. The controller dynamically computes available slots and updates a local OLED display.

The design ensures accurate occupancy tracking and reduces manual supervision within parking facilities. Slot data is also transmitted to the cloud API for remote visualization.

C. Entry Gate Control Module

The entry subsystem employs ultrasonic sensors for vehicle detection and a servo motor for barrier actuation. A finite state machine (FSM) governs system transitions between operational states, ensuring deterministic gate control.

The state transitions are defined as:

- STOP: Barrier closed, no vehicle detected
- SCAN: Vehicle detected, awaiting authorization
- GO: Barrier opened for vehicle entry

Traffic signal LEDs assist in safe vehicle movement. The FSM architecture prevents race conditions and ensures controlled operation.

D. Exit Safety Monitoring Module

To enhance pedestrian safety, a passive infrared (PIR) sensor monitors motion near the exit area. Upon detecting movement, a caution message is displayed to alert drivers.

This module operates continuously and independently of the entry system, ensuring localized safety monitoring without affecting overall system stability.

E. Maintenance and Hazard Detection Module

The maintenance subsystem integrates the moisture sensors for environmental risk detection. A priority-based control mechanism is developed.

The hazard states are classified as:

- Wet Surface Detection
- Normal Condition

Alerts are displayed using an LED matrix module to ensure high visibility. The prioritized design ensures immediate response to critical safety conditions.

F. Cloud API and Web Dashboard Integration

A cloud-based API and web dashboard are integrated to provide real-time system monitoring and slot management capabilities. The embedded controller periodically updates slot availability and system status to the cloud server.

The cloud platform enables:

- Remote slot status visualization
- Reservation management
- System monitoring
- Data accessibility through web interface

This integration extends the system beyond localized embedded control to a scalable smart infrastructure model.

6. System Overview

The proposed system is a modular smart parking infrastructure integrating real-time slot monitoring, automated vehicle entry control, pedestrian safety detection, environmental hazard monitoring, and cloud-based remote management. The architecture combines embedded sensing units, deterministic control logic, actuation mechanisms, and a web-enabled cloud interface to ensure operational efficiency and safety within parking environments.

The system is structured into four functional subsystems coordinated by embedded microcontroller units:

1. **Parking Slot Management System**
2. **Entry Gate Control System**
3. **Exit Safety Monitoring System**
4. **Maintenance and Hazard Detection System**

Each subsystem operates autonomously while contributing to the overall parking ecosystem.

The **Parking Slot Management System** performs real-time occupancy detection using infrared sensors. Slot availability is dynamically computed and displayed locally through an OLED interface. The system categorizes each slot into predefined logical states, ensuring accurate tracking of parking capacity.

The **Entry Gate Control System** regulates vehicle inflow using ultrasonic sensors and a servo-driven barrier mechanism. A finite state machine (FSM) governs the transition between operational states to ensure deterministic and safe gate operation. Traffic indication LEDs provide visual guidance to drivers during entry.

The **Exit Safety Monitoring System** enhances pedestrian safety by utilizing a passive infrared sensor to detect motion near the exit area. Upon detection, driver warning messages are activated to promote cautious vehicle movement.

The **Maintenance and Hazard Detection System** continuously monitors environmental conditions using different sensors. Hazard notifications are displayed through a high-visibility LED matrix module.

To extend system accessibility, a **Cloud API and Web Dashboard** have been developed for remote monitoring and management. The cloud interface enables real-time visualization of parking availability, reservation handling, and system status monitoring. This integration transforms the standalone embedded system into a scalable smart infrastructure model.

The overall system design emphasizes modularity, deterministic control through state-machine implementation, safety prioritization, and remote accessibility. This structured approach ensures reliability, maintainability, and practical feasibility for deployment in controlled parking environments.

7. Working Principle

The proposed smart parking system operates through coordinated sensing, state-based control, and cloud-enabled monitoring. Each subsystem functions independently while contributing to the overall parking infrastructure.

A. Parking Slot Monitoring

Infrared sensors detect vehicle presence in each parking slot. Based on sensor input, slots are classified as Free, Reserved, or Occupied. The microcontroller continuously updates slot availability and displays the free-slot count on an OLED module. The same data is transmitted to the cloud API for remote visualization through the web dashboard.

B. Entry Gate Control

Vehicle detection at the entrance is performed using ultrasonic sensors. A finite state machine governs barrier operation through three states: STOP, SCAN, and GO. Upon vehicle detection and authorization, a servo motor actuates the gate. The barrier automatically closes once the vehicle clears the entry zone, ensuring controlled and deterministic operation.

C. Exit Safety Monitoring

A PIR sensor monitors pedestrian motion near the exit area. When movement is detected, a caution message is displayed to alert drivers. In the absence of motion, the system remains in monitoring mode.

D. Maintenance and Hazard Detection

Sensors continuously monitor environmental conditions. A priority-based logic mechanism ensures the detection overrides other alerts. Hazard notifications are displayed using an LED matrix for high visibility.

E. Cloud Integration

The embedded controller periodically updates system status to a cloud-based API. The web dashboard provides real-time slot availability and system monitoring, extending the system beyond local hardware control.

8. Hardware

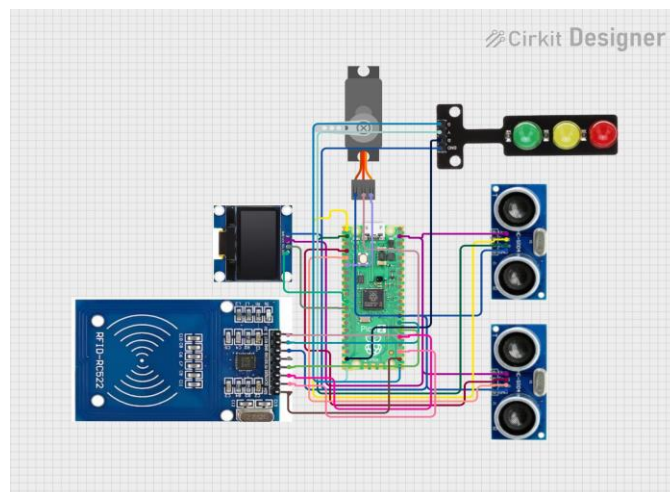


Fig.2 Entry Management System

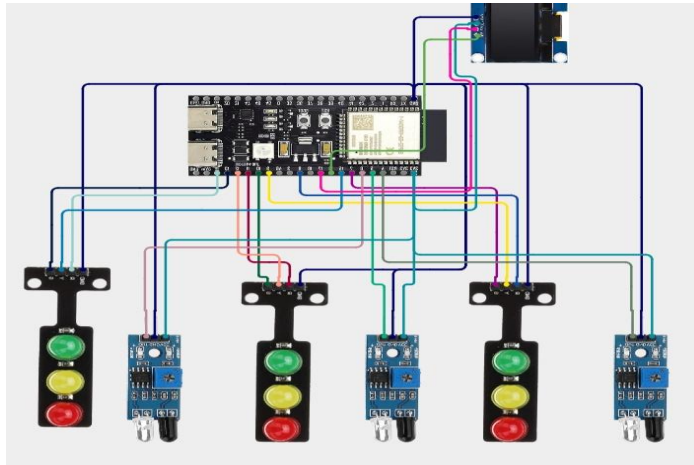


Fig.3 Parking Management System

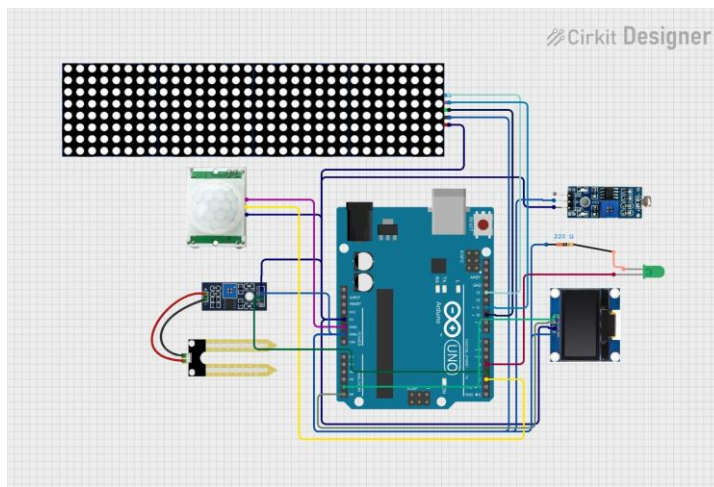


Fig.4 Exit and Hazard Management System

Microcontrollers

- **ESP32** – Parking slot management, WiFi communication, cloud API integration, OLED interface control
- **Raspberry Pi Pico (RP2040)** – Entry gate control, ultrasonic processing, servo actuation
- **Arduino UNO** – Exit safety monitoring, RFID authentication, LED matrix control

Sensors

- **Infrared (IR) Sensors** – Parking slot occupancy detection
- **Ultrasonic Sensor (HC-SR04)** – Vehicle detection at entry barrier
- **PIR Motion Sensor** – Pedestrian detection near exit zone
- **RFID Module (MFRC522)** – Vehicle/user authentication at controlled access points
- **Moisture Sensor Module** – Wet floor detection
- **LDR (Light Dependent Resistor)** – Ambient light monitoring and abnormal light variation detection

Actuators

- **Servo Motor** – Entry barrier gate control
- **LED Indicators (Red, Yellow, Green)** – Traffic and slot status indication
- **Buzzer** – Audible alert during hazard or authentication events

Display Units

- **0.96" OLED Display (SSD1306)** – Real-time parking availability and system information
- **8×8 LED Matrix (MAX7219)** – Hazard notifications (Wet Floor / Caution Alerts)

Communication & Networking

- **WiFi Module (ESP32 Built-in)** – Cloud API communication and real-time dashboard update
- **Web-Based Dashboard** – Remote slot monitoring and reservation management

Power & Supporting Components

- 5V Regulated Power Supply
- 3.3V Logic Level Support
- Breadboard
- Jumper Wires (Male–Male, Male–Female)
- Push Button (Manual Reset / System Control)

9. Software Architecture

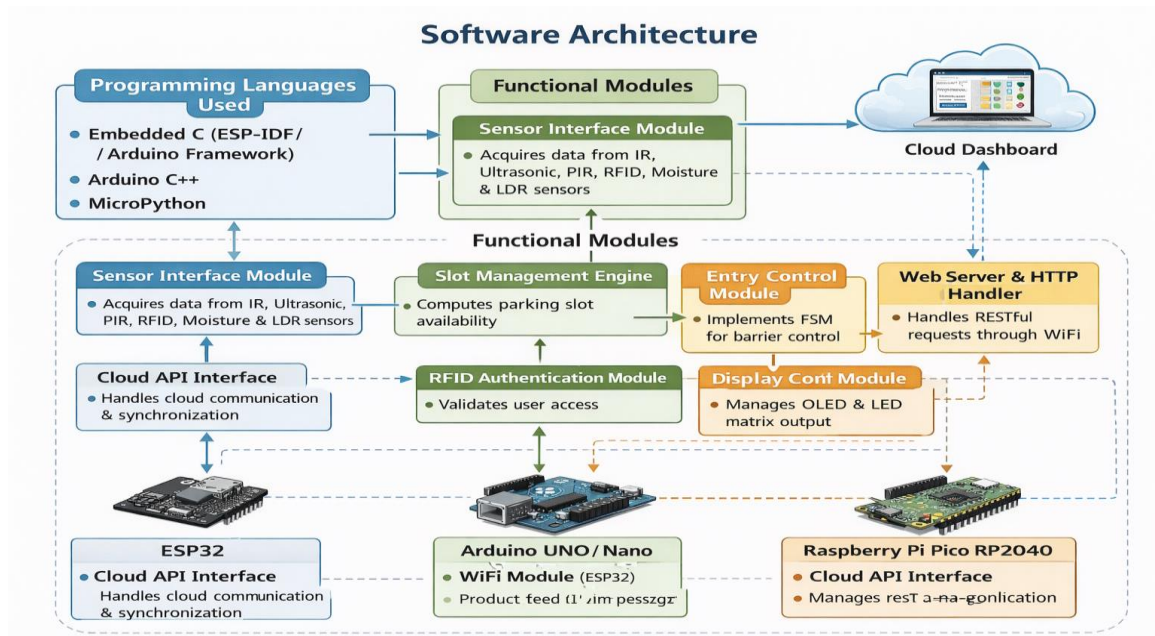


Fig.5 Software Architecture

The software architecture of the proposed smart parking system is designed using a modular and layered approach to ensure scalability, maintainability, and deterministic operation. The system integrates embedded firmware across multiple microcontrollers along with a cloud-based web interface for remote monitoring and management.

A. Programming Languages Used

- **Arduino C++** – Parking slot management, RFID authentication, LED matrix control, and exit monitoring using Arduino UNO
- **MicroPython** – Entry gate control (servo and ultrasonic processing) and embedded logic implementation on Raspberry Pi Pico
- **Embedded C (ESP-IDF / Arduino Framework for ESP32)** – Cloud communication, REST API handling, WiFi connectivity, and web dashboard integration

B. Functional Modules

• Sensor Interface Module

Handles acquisition of data from IR sensors, ultrasonic sensors, PIR sensor, RFID module, moisture sensor, and LDR. Performs threshold-based validation and filtering.

• Slot Management Engine

Processes parking occupancy states (Free, Reserved, Occupied) and computes real-time availability.

• Entry Control Module

Implements finite state machine logic (STOP–WAIT–GO) for deterministic barrier operation.

• RFID Authentication Module

Validates authorized users for controlled access and triggers servo actuation.

• Hazard Detection Engine

Implements priority-based evaluation of moisture and abnormal light conditions for safety alert generation.

• Display Control Module

Manages OLED and LED matrix visualization for slot status and hazard notifications.

• Web Server & HTTP Handler

Implements REST-based communication between ESP32 and cloud dashboard for remote monitoring and reservation management.

• Cloud API Interface

Handles data transmission, status updates, and real-time synchronization with the web platform.

The modular architecture ensures clear separation of concerns, deterministic control through state-machine implementation, and seamless integration between embedded systems and cloud infrastructure.

10. Bill of Materials

S.No	Component	Specification	Qty	Approx. Cost (₹)	Total (₹)
1	Raspberry Pi Pico	RP2040	1	350	350
2	ESP32 Dev Board	WiFi + Bluetooth	1	500	500
3	Arduino UNO	ATmega328P	1	600	600
4	IR Sensors	Slot Detection	3	80	240
5	Ultrasonic Sensor	HC-SR04	1	120	120
6	PIR Motion Sensor	HC-SR501	1	150	150
7	RFID Module	MFRC522	1	250	250
8	Moisture Sensor	Analog/Digital Module	1	120	120
9	LDR Sensor	Light Dependent Resistor	1	20	20
10	Servo Motor	SG90 / Equivalent	1	250	250
11	LED Matrix	MAX7219 8×8	1	300	300
12	OLED Display	0.96" SSD1306	3	250	750
13	LED Indicators	Red, Yellow, Green Set	3 Set	100	300
14	Buzzer	Piezo Buzzer	1	50	50
15	Power Supply	5V Regulated Adapter	1	300	300

11. Results & Discussions

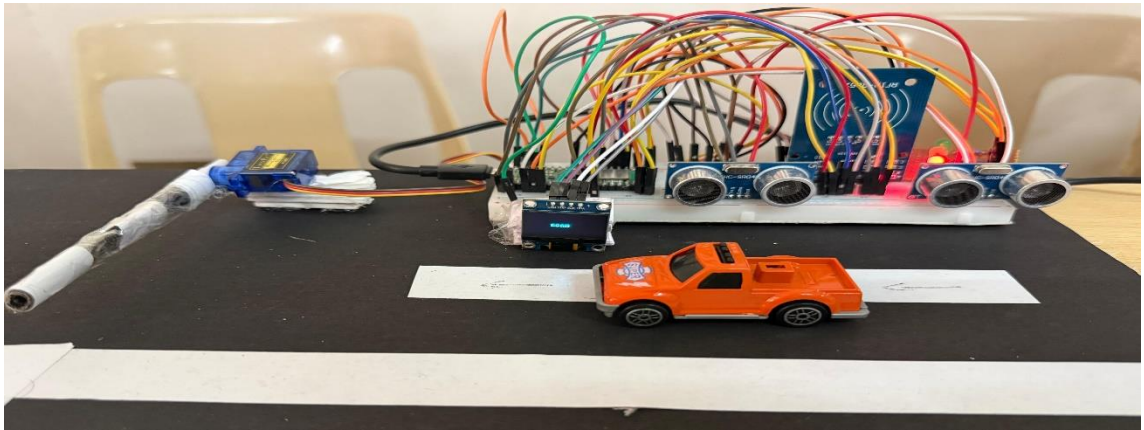


Fig.6 Vehicle detection at entry and RFID scan request by the system.

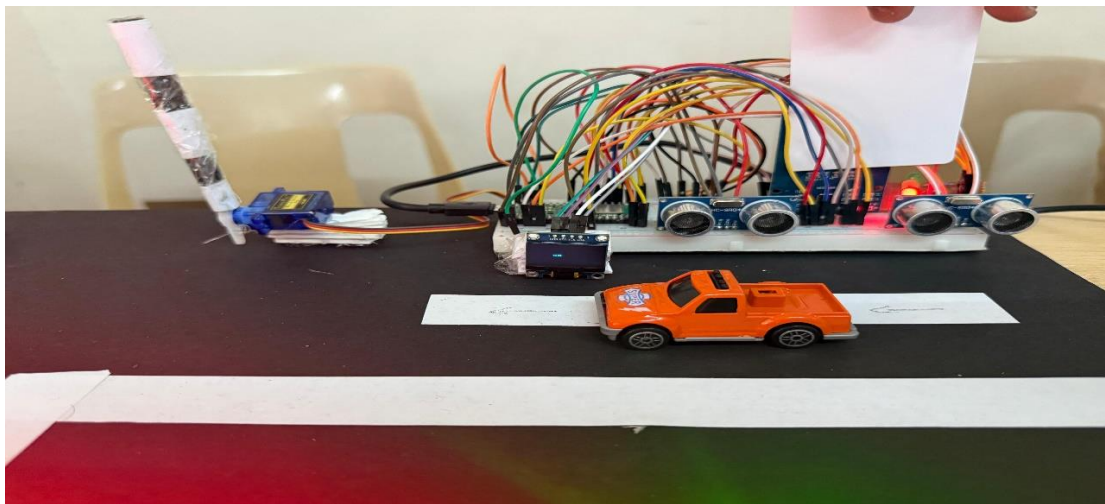


Fig.7 Vehicle detection at entry and RFID scan request by the system.

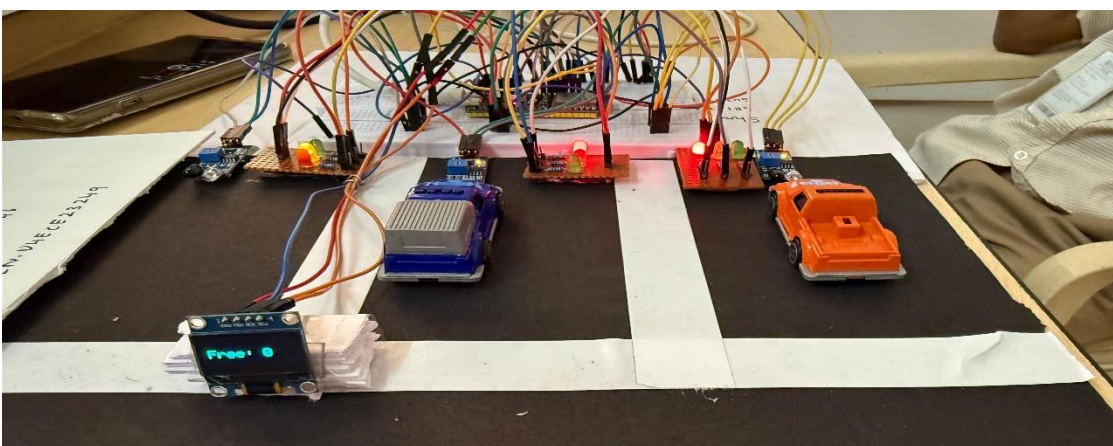


Fig.8 Parking slot indication showing two occupied slots (red LED) and one cloud-reserved slot (yellow LED).

Smart Parking System

Real-time monitoring and booking

[Manage Bookings →](#)

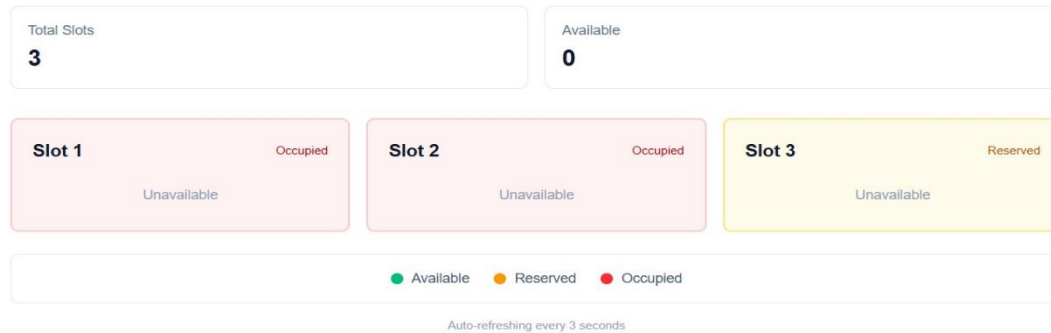


Fig.9 Web dashboard showing two occupied slots and one cloud-reserved slot

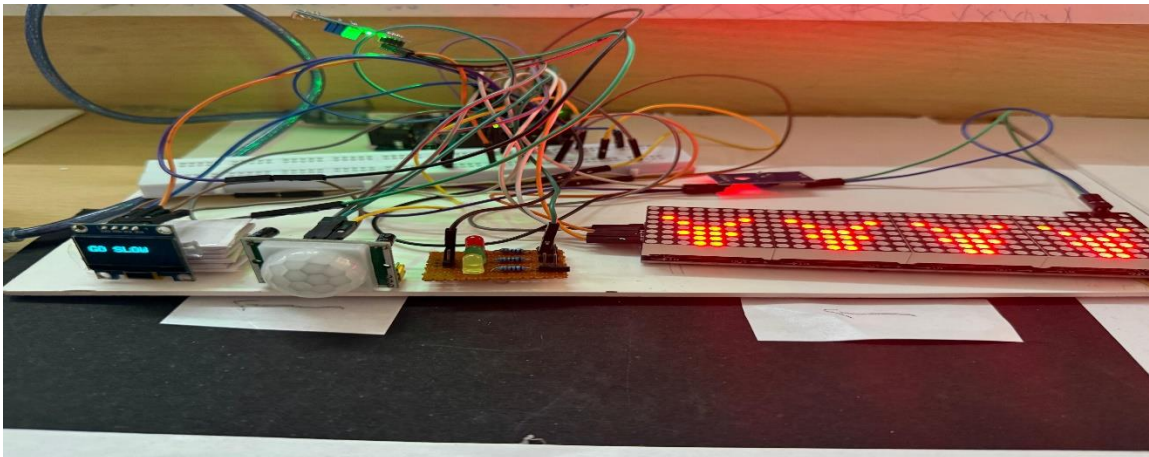


Fig.10 Pedestrian crossing indication on LED matrix with slow-down message on OLED display.

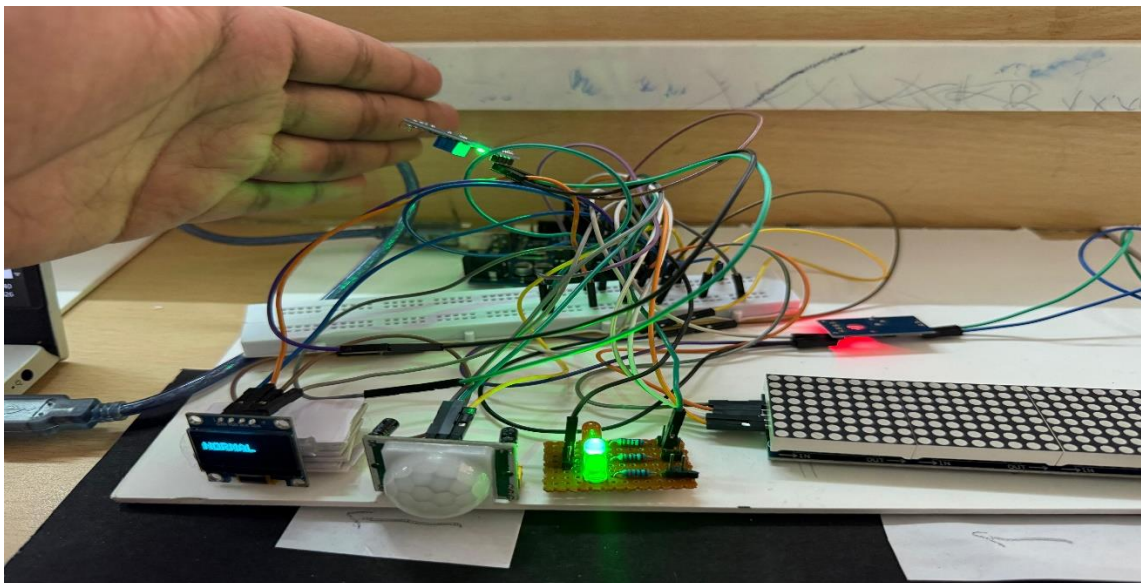


Fig.11 Ambient Light Detection using LDR.

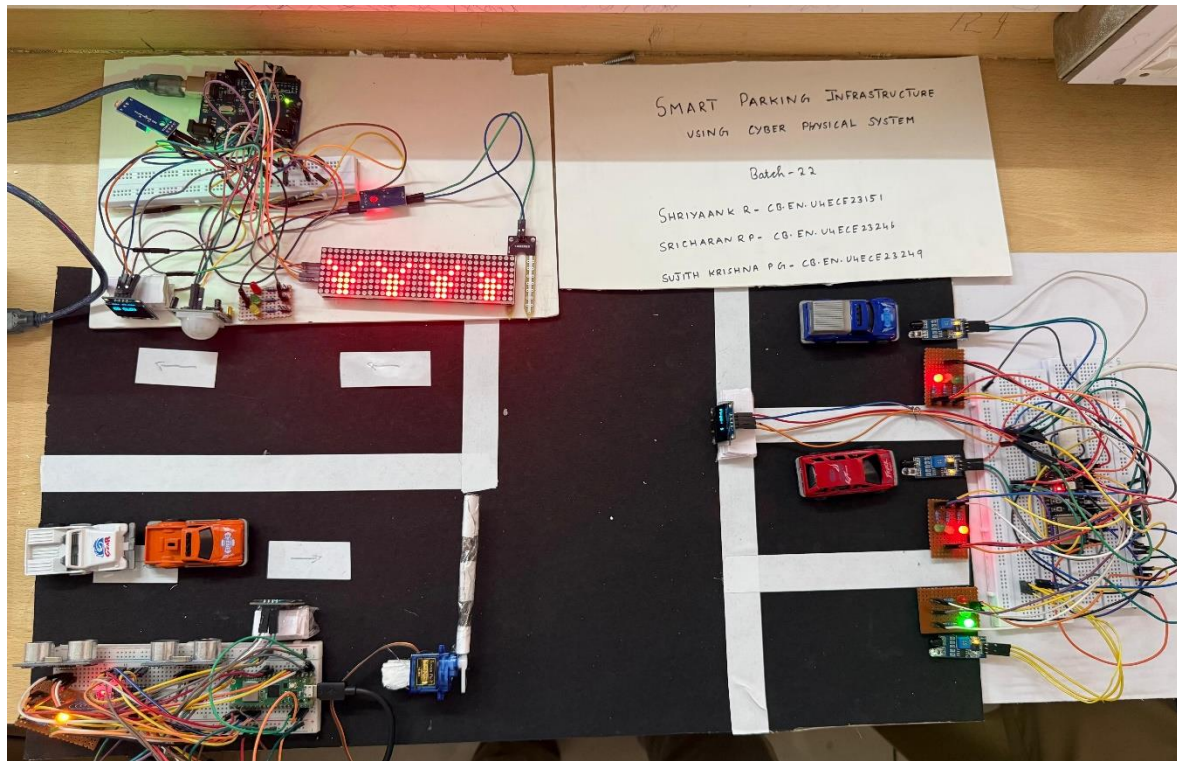


Fig.12 Overall Parking Infrastructure System

The developed smart parking system was tested under controlled conditions to evaluate sensing accuracy, control stability, and cloud synchronization performance.

The parking slot module accurately detected vehicle occupancy using IR sensors and displayed real-time availability without noticeable delay. The entry control system, governed by a finite state machine, demonstrated deterministic gate operation with reliable ultrasonic detection and smooth servo actuation. RFID authentication successfully validated authorized users and restricted unauthorized access.

The exit safety module effectively detected pedestrian movement and triggered caution alerts without false activation. The maintenance module accurately identified moisture conditions and displayed hazard notifications through the LED matrix. The priority-based logic ensured stable and conflict-free alert handling.

Cloud integration via ESP32 enabled consistent real-time updates to the web dashboard, confirming successful synchronization between embedded hardware and the remote platform.

Overall, the system achieved reliable sensing, controlled access management, prioritized safety handling, and seamless cloud connectivity, demonstrating practical feasibility for smart parking applications.

12.Future Scope and Applications

The proposed Cyber-Physical Smart Parking and Access Control System presents significant potential for future enhancement and large-scale deployment across smart infrastructure environments. The system can be extended through integration with cloud-based platforms and Internet of Things (IoT) technologies to enable centralized monitoring, real-time data analytics, and mobile-based supervision. Advanced features such as camera-based vehicle identification, automatic number plate recognition (ANPR), and artificial intelligence-driven parking optimization algorithms can further improve automation accuracy and operational efficiency. Additionally, the incorporation of renewable energy sources such as solar power can enhance sustainability, while expansion to multi-level parking facilities and dynamic pricing mechanisms can support commercial scalability.

The system is applicable in a wide range of environments requiring efficient parking management and secure vehicle access control. It can be deployed in commercial complexes, corporate offices, educational institutions, residential communities, hospitals, transportation hubs, government buildings, and smart city infrastructure projects. By reducing congestion, improving security, and optimizing space utilization, the proposed solution contributes to sustainable urban mobility and intelligent transportation management systems.

13. Project Outcomes

The implementation of the proposed Cyber-Physical Smart Parking and Access Control System is expected to achieve the following outcomes:

1. Successful development of an embedded system capable of real-time vehicle detection and parking slot monitoring.
2. Implementation of an automated gate access control mechanism ensuring regulated vehicle entry and exit.
3. Integration of multi-sensor security monitoring for enhanced safety and intrusion detection.
4. Real-time display of parking availability and system status through user interface modules.
5. Wireless communication capability enabling remote supervision and monitoring.
6. Demonstration of a functional Cyber-Physical System integrating sensing, processing, and actuation within a unified framework.
7. Improved parking space utilization leading to reduced congestion and operational inefficiencies.
8. Validation of a scalable, energy-efficient, and cost-effective solution suitable for smart city infrastructure applications.

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CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

Title of the Project : **Smart Parking Infrastructure Using Cyber-Physical Systems**

Project Member(s) Name & Register Number :

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Guide Name & Designation :

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Coimbatore**

SDGs	Justification to the Contribution
9	The system supports SDG 9 (Industry, Innovation and Infrastructure) by implementing an embedded, sensor-based smart infrastructure framework that integrates real-time sensing, automated control, and wireless communication, thereby fostering innovation in urban facility management.
11	It contributes to SDG 11 (Sustainable Cities and Communities) by reducing traffic congestion and improving parking efficiency through intelligent slot monitoring and automated gate management, ultimately enabling safer and more efficient urban environments.
12	The project aligns with SDG 12 (Responsible Consumption and Production) by optimizing parking space utilization, minimizing unnecessary vehicle circulation, and reducing fuel wastage through data-driven decision-making.
13	Furthermore, it supports SDG 13 (Climate Action) by decreasing idle driving time and associated carbon emissions, while encouraging energy-efficient embedded system deployment within smart transportation infrastructure.

1 NO POVERTY



2 ZERO HUNGER



3 GOOD HEALTH AND WELL-BEING



4 QUALITY EDUCATION



5 GENDER EQUALITY



6 CLEAN WATER AND SANITATION



7 AFFORDABLE AND CLEAN ENERGY



8 DECENT WORK AND ECONOMIC GROWTH



9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



10 REDUCED INEQUALITIES



11 SUSTAINABLE CITIES AND COMMUNITIES



12 RESPONSIBLE CONSUMPTION AND PRODUCTION



13 CLIMATE ACTION



14 LIFE BELOW WATER



15 LIFE ON LAND



16 PEACE, JUSTICE AND STRONG INSTITUTIONS



17 PARTNERSHIPS FOR THE GOALS

